

 **INPUT** TLE catalog + maneuver screening

1



Layer 1 — Physical safety shell

SGP4 baseline; fallback on gate rejection

2



Layer 2 — AI enhancement layer

Transformer / Bi-LSTM ensemble; learns residual $\hat{R}$

train

3



Layer 3 — Fail-Safe gate

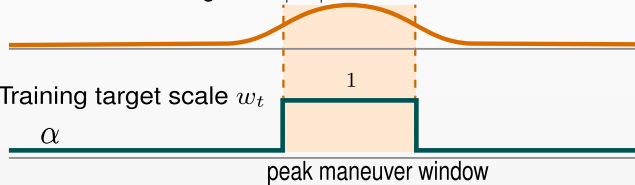
Falls back to SGP4 when unreliable



OUTPUT corrected orbit + risk-gated decision

Core innovation: Soft-Mask training

SGP4 residual magnitude $|R_t|$



α = non-peak target-retention factor

$$\ell(\theta) = \frac{1}{T} \sum_t \|\hat{R}_t - w_t R_t\|^2$$

$w_t = 1$ on peak windows, $w_t = \alpha$ elsewhere

Physical prior: residual deviation is spatially sparse
peak-window mask suppresses off-peak AI noise
 t_{peak} : used for training only

Evaluation metrics (SGP4 baseline $\rightarrow$ proposed; large-maneuver group, $n = 817$)

Superior-to-SGP4

SGP4 baseline
65.5%

proposed
82.7%

Severe degradation ($\eta < -20\%$)

SGP4 baseline
9.3%

proposed
3.3%

Worst-case degradation

SGP4 baseline
-199.8%

proposed
-77.6%

Non-peak AI noise

SGP4 baseline
 $1.40 \times 10^8 \text{ km}^2$

proposed
 $2.55 \times 10^7 \text{ km}^2$

Conceptual basis: bias–variance decomposition · Stein-type shrinkage (analogy) · stress–strength interference (illustrative)